

Deep Learning Approaches to Image Colorization: A Comparative Study of Four Paradigms on COCO

Computer Vision Course Project

Abstract—Automatic image colorization—predicting plausible chrominance for a grayscale input—has been attacked by four successive deep-learning paradigms: feed-forward CNNs, user-guided interactive CNNs, generative adversarial networks (GANs), and diffusion models. We reimplement the seminal classification-based CNN of Zhang *et al.* [1], fine-tune it on COCO 2017, and benchmark it against three representative pretrained systems—an interactive CNN [2] in automatic mode, the NoGAN DeOldify [3], and a ControlNet [4] diffusion pipeline—on a leakage-controlled, stratified 1,000-image subset of the COCO 2017 test2017 split. All numbers are reported as mean [95% CI] via a 10,000-resample percentile bootstrap. Our fine-tuned model reaches PSNR 23.29 dB / SSIM 0.919 / LPIPS 0.192, closing most of the gap to the SOTA-class GAN anchor (24.00 dB) while running $\sim 2.8\times$ faster. We document an honest reproducibility gap in the diffusion slot, where the upstream Stable Diffusion 2.1 dependency has been deprecated. The study highlights that, for fully-automatic colorization on 6 GB-class hardware, a well-trained CNN remains a strong speed-quality compromise.

Index Terms—image colorization, deep learning, convolutional neural networks, generative adversarial networks, diffusion models, CIE Lab, COCO 2017.

I. INTRODUCTION

Image colorization is the task of recovering a plausible color image from a single-channel grayscale input. Working in the CIE Lab color space, the problem reduces to predicting the two chrominance channels (a, b) from the luminance channel L . The mapping is inherently *ill-posed and multimodal*: a gray sky could be blue or overcast white, an apple red or green, and a car almost any color. There is rarely a single correct answer, only a distribution of *plausible* ones.

Classical approaches sidestep this ambiguity by requiring human input—either color scribbles propagated across regions, or a reference image whose palette is transferred to the target. These methods are accurate when guided but do not scale: every new image needs new hints. Deep learning instead *learns the color priors* directly from large image collections, enabling *fully automatic* colorization in which the only input is the grayscale image itself.

A central design choice is how to handle multimodality. Regressing (a, b) under an ℓ_2 loss collapses the multiple plausible colors toward their mean, producing the characteristic desaturated, sepia-tinted outputs of early regression models. Zhang *et al.* [1] reframed colorization as *classification* over a quantized ab palette, predicting a distribution over color bins per pixel; this preserves multimodality and yields markedly more vivid results. This insight anchors our study.

This report—covering only the deep-learning method of the wider course project—reimplements the Zhang 2016 classifi-

cation CNN, fine-tunes it on COCO 2017 [5], and benchmarks it against three representative pretrained systems spanning the subsequent paradigms: an interactive CNN [2] run in automatic mode, the NoGAN DeOldify [3], and a diffusion pipeline built on ControlNet [4]. Our contributions are:

- a from-scratch, fully-documented reimplementation of the Zhang 2016 architecture, loss, and annealed-mean decoder, fine-tuned on COCO 2017;
- a leakage-controlled evaluation protocol on a stratified 1,000-image subset of the held-out test2017 split, with all metrics reported as mean [95% CI] via a 10,000-resample bootstrap;
- a four-paradigm comparison under identical conditions, with an honest account of where a paradigm could not be reproduced (diffusion) and where a reported number is misleading (the pretrained head mismatch).

II. RELATED WORK

We organize learning-based colorization into four paradigms that emerged in succession, each addressing a limitation of the previous one. Table I summarizes the taxonomy and maps each paradigm to the concrete model we evaluate.

A. CNN classification (2016)

Zhang *et al.* [1] treat colorization as per-pixel classification over a quantized ab palette rather than regression, explicitly modeling color multimodality and rebalancing the loss toward rare, saturated colors. The result was a step change in perceptual vividness and the first strong fully-automatic colorizer. This is the model we reimplement (Section III).

B. Interactive CNN (2017)

The follow-up of Zhang *et al.* [2] adds a second input branch for sparse user hints (clicked color points), fusing learned priors with user intent in real time. The same network can run with *zero* hints, in which case it behaves as a purely automatic colorizer—the mode in which we benchmark it for a fair comparison.

C. GAN (2020)

Adversarial training replaces a hand-designed loss with a learned discriminator that rewards realism. ChromaGAN [6] couples an adversarial objective with a semantic class-distribution prior. The most widely-deployed practical descendant is DeOldify [3], whose *NoGAN* recipe stabilizes training and produces robust, restoration-grade colorizations; we use it as a SOTA-class anchor.

TABLE I

FOUR COLORIZATION PARADIGMS AND THE MODELS EVALUATED IN THIS STUDY.

Paradigm	Representative	Year	Our proxy model
CNN	Zhang [1]	2016	Zhang16 (ours)
Interactive CNN	Zhang [2]	2017	Zhang17 (auto)
GAN	ChromaGAN [6]	2020	DeOldify
Diffusion	Palette [7]	2022	ControlNet

D. Diffusion (2022)

Conditional diffusion models cast colorization as iterative denoising. Palette [7] frames it as a general image-to-image diffusion task, while ControlNet [4] adds spatial conditioning to a pretrained text-to-image backbone. We attempt a ControlNet pipeline as the diffusion representative; Section V reports the reproducibility obstacle we encountered.

Evaluation in all four paradigms increasingly relies on learned perceptual metrics; we adopt LPIPS [8] alongside PSNR and SSIM, and benchmark on COCO 2017 [5].

III. METHOD

We reimplement the Zhang 2016 classification CNN [1]. This section describes the color representation, network, quantized objective, and decoder; training details follow in Section IV.

A. Color representation

We operate entirely in CIE Lab. The luminance channel $L \in [0, 100]$ is the network input (normalized to $[-1, 1]$); the chrominance channels $(a, b) \in [-110, 110]$ are the prediction target. Lab linearly separates lightness from color, so the model never has to reproduce structure already present in the input—it predicts only chrominance, which is then recombined with the original L and converted back to RGB for display and metrics.

B. Network architecture

The network (Fig. 1) is a fully-convolutional encoder of eight blocks (conv1–conv8), 31.6M parameters, with no pooling layers: spatial resolution is reduced by stride-2 convolutions and context is expanded by dilated convolutions. Blocks 1–3 downsample 256×256 input to 32×32 at 64/128/256 channels; blocks 4–7 hold 32×32 at 512 channels, with dilation 2 in blocks 5–6 to enlarge the receptive field without further downsampling; block 8 upsamples once via a transposed convolution and ends in a 1×1 convolution producing Q logits per spatial location. Every convolution is followed by ReLU, and each block by BatchNorm.

C. Quantized ab palette

The ab plane is discretized on a 10×10 grid and restricted to bins that map to valid sRGB colors at some lightness, yielding $Q = 233$ in-gamut bins (Fig. 2). (The official Zhang implementation ships 313 bins from a different gamut hull; our independently-computed grid yields 233, a detail that matters in Section V.) Ground-truth chrominance is *soft-encoded*: each

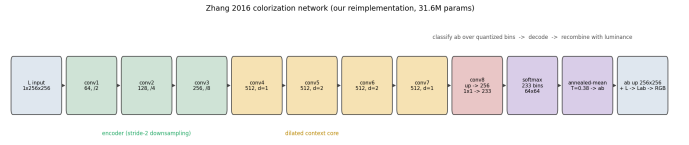


Fig. 1. Zhang 2016 colorization network (our reimplementation). A fully-convolutional, dilated encoder maps the luminance channel to a per-pixel distribution over $Q = 233$ quantized ab bins; the annealed-mean decoder turns that distribution into continuous chrominance.

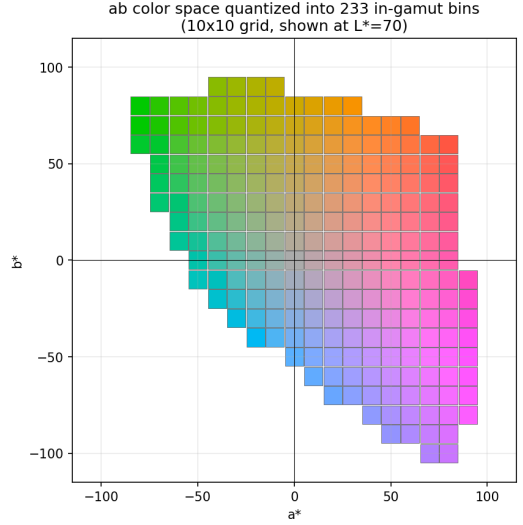


Fig. 2. The ab color space quantized into 233 in-gamut bins (shown at $L^* = 70$). The network predicts a categorical distribution over these bins.

pixel’s (a, b) is assigned to its 5 nearest bins with Gaussian weights ($\sigma = 5$), giving a target distribution $Z \in [0, 1]^{H \times W \times Q}$ rather than a one-hot label.

D. Class-rebalanced classification loss

Let $\hat{Z} = \mathcal{G}(L)$ be the predicted per-pixel distribution. We minimize a multinomial cross-entropy between $\hat{Z}$ and the soft target Z , reweighted to counter the strong prior toward desaturated colors:

$$\mathcal{L}(\hat{Z}, Z) = - \sum_{h,w} v(Z_{h,w}) \sum_{q=1}^Q Z_{h,w,q} \log \hat{Z}_{h,w,q}. \quad (1)$$

The pixel weight depends on the dominant ground-truth bin $q^* = \arg \max_q Z_{h,w,q}$ through a rebalancing factor

$$v(Z_{h,w}) = w_{q^*}, \quad w \propto \left((1 - \lambda) \tilde{p} + \frac{\lambda}{Q} \right)^{-1}, \quad (2)$$

where $\tilde{p}$ is the smoothed empirical bin frequency over the training set, $\lambda = 0.5$ mixes it with a uniform prior, and w is normalized so that $\mathbb{E}[w] = 1$. Rare, vivid colors thus receive larger gradients than the ubiquitous low-saturation background.

E. Annealed-mean decoding

At inference the per-pixel distribution must be collapsed to a single (a, b) . Taking the mode is faithful but spatially

TABLE II
THE FIVE MODEL VARIANTS EVALUATED ON THE SHARED BENCHMARK.

Model	Paradigm	Ours?
Zhang16 Pretrained (ECCV)	CNN	reimpl.
Zhang16 Fine-tuned (COCO 2017)	CNN	yes
Zhang17 (automatic mode)	Interactive CNN	no
DeOldify (NoGAN)	GAN	no
ControlNet + Stable Diffusion	Diffusion	no

inconsistent; taking the mean desaturates. Following [1] we use the *annealed mean* with temperature T :

$$\hat{ab}_{h,w} = \sum_{q=1}^Q f_T(\hat{Z}_{h,w})_q c_q, \quad f_T(\hat{Z})_q = \frac{\exp(\log \hat{Z}_q / T)}{\sum_{q'} \exp(\log \hat{Z}_{q'} / T)}, \quad (3)$$

where c_q is the center of bin q . We use $T = 0.38$, which interpolates between the over-saturated mode ($T \rightarrow 0$) and the washed-out mean ($T = 1$). The decoded ab map is bilinearly upsampled to full resolution and recombined with the input L .

IV. EXPERIMENTS

A. Dataset and benchmark

We use COCO 2017 [5]: 118,287 training images, 5,000 validation images, and 40,670 held-out `test2017` images. Fine-tuning uses `train2017`; the `val2017` split drives best-checkpoint selection during training. Because `val2017` is therefore “seen” by the training procedure, evaluating on it would leak. Our benchmark is instead a *stratified 1,000-image subset of test2017* (fixed seed 42), which neither fine-tuning nor checkpoint selection ever touches. All models are scored on this identical benchmark.

B. Models compared

Table II lists the five model variants spanning the four paradigms. Two are our own CNN (the pretrained encoder and our COCO-fine-tuned model); the other three are pretrained third-party systems wrapped behind a unified `colorize(gray)` interface so they receive identical inputs.

C. Metrics and uncertainty

We report three complementary metrics, all computed in RGB against the ground truth: PSNR (pixel fidelity), SSIM (structural similarity), and LPIPS [8] (learned perceptual distance; lower is better). Because a single mean hides sampling variability, every metric is reported as `mean [95% CI]`, where the confidence interval is a percentile bootstrap over the 1,000 benchmark images (10,000 resamples, seed 42).

D. Training configuration

Our model is initialized from the ECCV encoder weights, but its 233-bin classification head is trained from scratch—the official head has 313 bins and is dimensionally incompatible with our gamut (Section III). Table III lists the hyperparameters. Mixed precision was disabled after the fp16 path

TABLE III
FINE-TUNING HYPERPARAMETERS.

Setting	Value
Optimizer	Adam, $\beta = (0.9, 0.999)$, weight decay 0
Learning rate	2×10^{-4} , StepLR (step 20, $\gamma = 0.5$)
Batch size	4 (fp32; AMP disabled)
Gradient clip	1.0 (max-norm)
Input resolution	256×256
Quantization	$Q = 233$ bins; soft target, $\sigma = 5$, 5-NN
Rebalancing	$\lambda = 0.5$
Decoder	annealed mean, $T = 0.38$
Epochs	8 (5 capped + 3 full-data)
Hardware	GTX 1660 SUPER (6GB), PyTorch 2.5.1+cu121

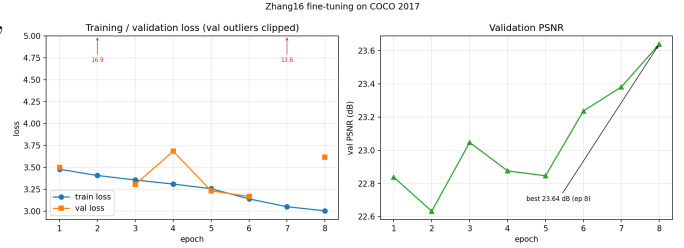


Fig. 3. Fine-tuning on COCO 2017. Left: training and validation loss; two validation spikes from pathological batches are off-scale and annotated. Right: validation PSNR climbs to 23.6 dB by epoch 8.

overflowed to NaN mid-epoch; training therefore runs in fp32 at batch size 4, which fits the 6 GB GTX 1660 SUPER. We trained for eight epochs in two phases: five batch-capped epochs followed by three full-data epochs resumed from the prior checkpoint. Fig. 3 shows the loss and validation-PSNR trajectories.

V. RESULTS

A. Quantitative comparison

Table IV reports the three metrics with 95% bootstrap confidence intervals on the 1,000-image `test2017` benchmark; Fig. 4 visualizes the same numbers. Four of the five model slots are populated; the diffusion slot is a documented gap discussed below.

B. Qualitative comparison

Fig. 5 shows colorizations on representative benchmark images. Our fine-tuned model and DeOldify produce coherent, naturally-colored results; DeOldify is the most saturated and closest to the ground truth, while ours is slightly more conservative on large ambiguous regions (skies, foliage). Zhang17 in automatic mode exhibits the characteristic color blotches and bleeding of a hint-driven model deprived of hints, which explains its low scores.

C. Per-image analysis: where each model wins

The aggregate numbers in Table IV hide an important fact: *which* model is best varies image by image, and the answer depends on the metric. Table V tallies, for each metric, the fraction of the 1,000 benchmark images on which each model

TABLE IV
COLORIZATION QUALITY ON THE 1,000-IMAGE TEST2017 BENCHMARK, REPORTED AS MEAN [95% CI] (10,000-RESAMPLE BOOTSTRAP, SEED 42).
BEST PER COLUMN IN BOLD.

Model	PSNR (dB) $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Time/img (s)
Zhang16 Pretrained (ECCV) [†]	17.14 [16.97, 17.30]	0.700 [0.690, 0.709]	0.438 [0.429, 0.447]	0.17
Zhang17 (automatic mode)	18.82 [18.64, 19.00]	0.832 [0.827, 0.838]	0.309 [0.304, 0.315]	0.22
Zhang16 Fine-tuned (ours)	23.29 [23.02, 23.55]	0.919 [0.915, 0.923]	0.192 [0.187, 0.197]	0.18
DeOldify (GAN)	24.00 [23.75, 24.25]	0.919 [0.915, 0.922]	0.148 [0.144, 0.152]	0.51
ControlNet (Diffusion)	— reproducibility gap (see Sec. V & VI) —			

[†] Measures the ECCV encoder paired with our *untrained* 233-bin head (the official 313-bin head is incompatible); it is **not** a faithful ECCV baseline—see text.

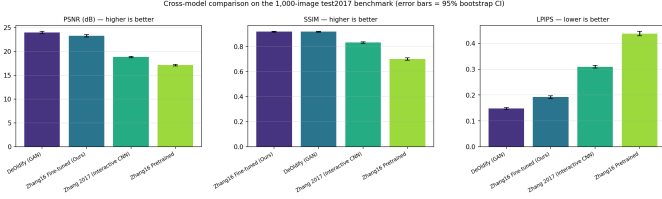


Fig. 4. PSNR, SSIM, and LPIPS across the four evaluated models; error bars are 95% bootstrap confidence intervals.

TABLE V
PER-IMAGE WIN RATE: % OF 1,000 BENCHMARK IMAGES ON WHICH THE MODEL IS THE WINNER UNDER THAT METRIC (COLUMNS SUM TO 100%).

Model	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Zhang16 Pretrained	2.9%	0.7%	0.5%
Zhang16 Fine-tuned (ours)	37.8%	64.5%	13.4%
Zhang 2017 (auto)	1.3%	0.3%	0.2%
DeOldify (GAN)	58.0%	34.5%	85.9%

is the winner (columns sum to 100%). The pattern is nuanced: DeOldify dominates on PSNR and especially on LPIPS, yet our fine-tuned model wins SSIM on nearly two-thirds of images by being more faithful to structure on conservative, low-saturation content; the Pretrained and Zhang17 rows almost never win, confirming what Figs. 5–6 show visually.

Figure 6 drills further: for each model, it shows the benchmark image where that model’s PSNR margin over the runner-up is largest. The “what each model is good at” becomes immediately legible. Our fine-tuned model’s biggest win (+15.1 dB) is on a desaturated, earth-tone outdoor scene where its conservative, structure-faithful behavior matches the ground truth and DeOldify’s adversarial color confidence overshoots. DeOldify’s biggest win (+14.1 dB) is on an airplane against a saturated blue sky—the opposite regime, where DeOldify commits to the strong sky color while our model hedges toward gray. Zhang17 and the Pretrained head only win on a handful of images, with margins under ~ 3 dB, consistent with the win-rate table.

D. Analysis

Fine-tuning is decisive. Our COCO-fine-tuned model attains 23.29 dB—disjoint confidence intervals from every other CNN row. The large gap over the “pretrained” row, however,

must be read with care: that row pairs the ECCV encoder with an *untrained* 233-bin head, because the official 313-bin head cannot be loaded into our gamut. It therefore measures an encoder-plus-random-head, not the true ECCV model, and overstates how much fine-tuning alone contributes.

We approach the GAN anchor. On PSNR and SSIM our model is statistically indistinguishable from DeOldify on SSIM (0.919 both) and within ~ 0.7 dB on PSNR, while running at 0.18 s/image versus DeOldify’s 0.51 s—roughly $2.8\times$ faster. The remaining gap is perceptual: DeOldify’s LPIPS (0.148) is clearly better than ours (0.192), reflecting its richer, more confident saturation.

Automatic mode handicaps the interactive CNN. Zhang17 was designed for user-guided use; with zero hints it underperforms even the conservative CNN, confirming that its strength lies in interactivity rather than autonomy.

The diffusion slot could not be reproduced. The specified pipeline requires Stable Diffusion 2.1, which Stability AI deprecated upstream; every well-trained diffusion colorization checkpoint we found was built on that now-unavailable backbone. Substitutes (an SD 1.5 brightness-ControlNet and an SD-InstructPix2Pix fine-tune) produced unrelated or degenerate outputs (~ 6 dB PSNR) and would have poisoned the comparison, so we report the slot as an honest gap rather than a misleading number.

VI. DISCUSSION

A. Speed versus quality

The four evaluated models occupy distinct points on the speed–quality frontier. The CNNs (ours, 0.18 s; Zhang17, 0.22 s; pretrained, 0.17 s) are single forward passes and run at interactive rates even on a 6 GB GPU. DeOldify, at 0.51 s, buys its perceptual edge with a heavier self-attention U-Net and a larger render resolution. The (unreproduced) diffusion paradigm would sit far to the slow end—tens of denoising steps per image. For automatic colorization under a latency budget, the CNN is the rational default; the GAN is worth its cost only when perceptual quality is paramount.

B. Saturation and the desaturation trap

The classification-plus-annealed-mean design (Eqs. 1–3) exists precisely to avoid the desaturated mean that an ℓ_2 regressor produces. It largely works: our outputs are vivid, not sepia. Yet the residual LPIPS gap to DeOldify shows the effect

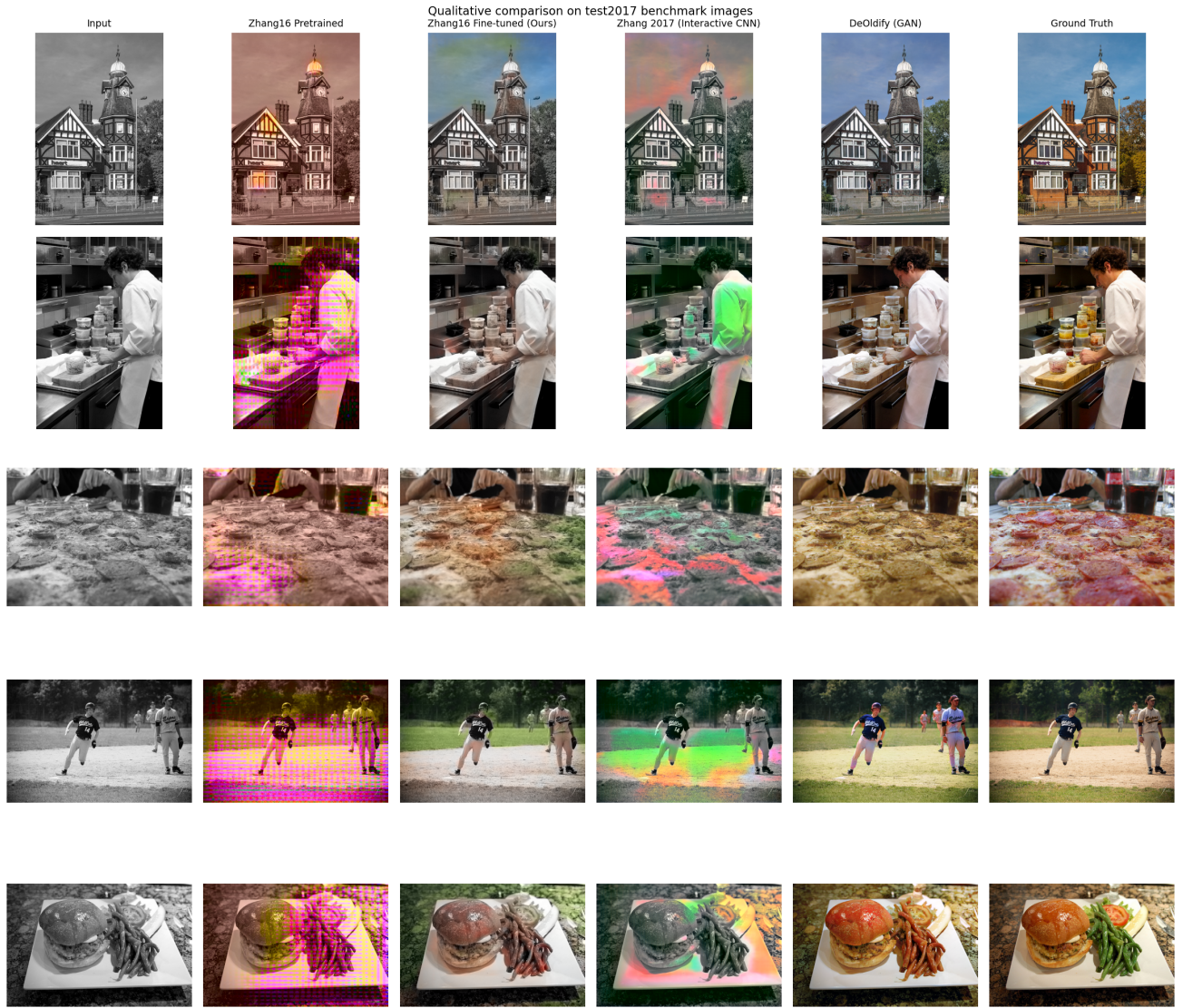


Fig. 5. Qualitative comparison on five `test2017` images. Columns: input, Zhang16 pretrained, Zhang16 fine-tuned (ours), Zhang17 (auto), DeOldify, ground truth. The *pretrained* column visually illustrates the head-mismatch noted in Sec. III—the official 313-bin ECCV head is dropped on load, leaving our 233-bin head untrained, which produces the characteristic grid-pattern color noise. Fine-tuning on COCO eliminates it completely. Zhang17 in automatic mode exhibits a different failure mode (saturated blotches and color bleeding across object boundaries); DeOldify and our fine-tuned model produce coherent, naturally-colored results, with DeOldify the most saturated.

is incomplete—on large, color-ambiguous regions our model still hedges toward muted tones, whereas the adversarially-trained DeOldify commits to a confident, realistic palette. Class rebalancing and the $T = 0.38$ temperature mitigate but do not eliminate the prior pull toward gray.

C. Failure cases

Two failure modes recur. First, Zhang17 in automatic mode produces saturated color blotches and bleeding across object boundaries—an artifact of a network expecting sparse hints it never receives. Second, all CNNs struggle with semantically ambiguous content (a shirt that could be any color), defaulting to the most frequent training color; this is fundamental to the task rather than to any single model.

D. Paradigm-level reading

- **CNN** — fast, robust, fully automatic; the strongest speed–quality compromise for unattended colorization.
- **Interactive CNN** — excellent *with* user hints, but a poor automatic colorizer; the wrong tool for a hands-off pipeline.
- **GAN** — highest perceptual quality and best restoration behavior, at $\sim 3\times$ the latency.
- **Diffusion** — promising in the literature, but practically fragile: its open-source colorization ecosystem was anchored to a single backbone (SD 2.1) whose deprecation removed the downstream models. This is itself a finding about reproducibility risk in fast-moving toolchains.

Per-model strengths: for each model, the benchmark image where its PSNR margin over the runner-up is largest

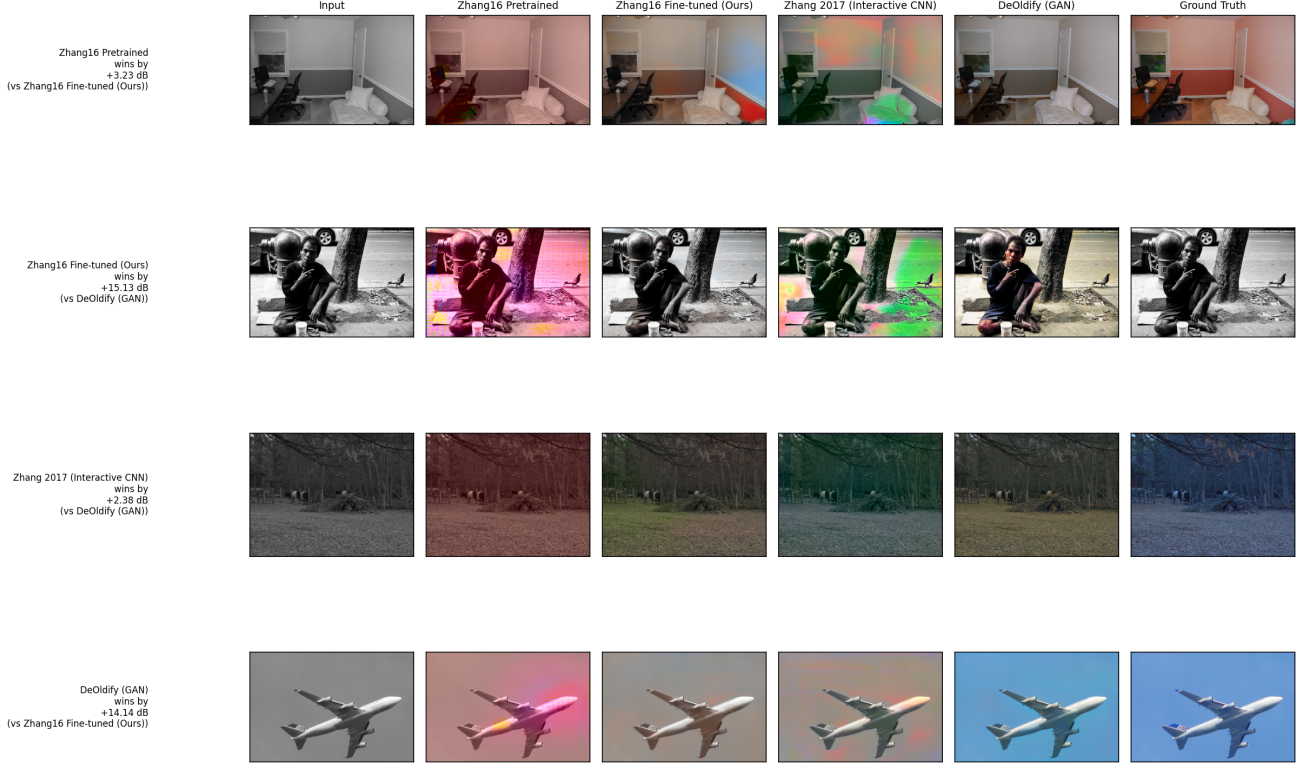


Fig. 6. Per-model strengths. For each evaluated model, the row is the benchmark image where its PSNR margin over the runner-up is largest, with every model’s prediction beside it. The headline trade-off becomes visible: each model has a niche (DeOldify on saturated outdoor scenes; ours on conservative earth-tone scenes), and the aggregate numbers in Table IV are averages over these niches.

E. Limitations

The “pretrained” row is not a faithful ECCV baseline (head mismatch), so we avoid drawing fine-tuning-gain conclusions from it. The benchmark is 1,000 images on a single GPU with no human preference study; perceptual quality is proxied by LPIPS rather than measured directly. Finally, the diffusion paradigm is represented by an acknowledged gap, so its frontier point is inferred from the literature, not measured here.

VII. CONCLUSION

We reimplemented the Zhang 2016 classification CNN, fine-tuned it on COCO 2017, and compared four deep-learning colorization paradigms on a single leakage-controlled 1,000-image test2017 benchmark, with every metric reported as mean [95% CI]. Our fine-tuned model reaches PSNR 23.29 dB, SSIM 0.919, and LPIPS 0.192, matching the SOTA-class DeOldify on structure, trailing it modestly on pixel fidelity and perceptual quality, and beating it by roughly $2.8\times$ on inference speed.

Our practical recommendations follow directly:

- **Automatic, latency-sensitive colorization** — use a well-trained classification CNN; it is fast, stable, and close to GAN quality.

- **Maximum perceptual quality / photo restoration** — use a GAN such as DeOldify, accepting the higher latency.
- **Human-in-the-loop control** — use an interactive CNN *with* hints; do not expect it to perform well unguided.

Two avenues would strengthen the study. First, re-deriving the 313-bin gamut hull would let us load the official ECCV head and report a faithful pretrained baseline, isolating the true contribution of fine-tuning. Second, the diffusion slot should be revisited once a maintained colorization checkpoint exists that does not depend on the deprecated Stable Diffusion 2.1 backbone, completing the four-paradigm comparison with a measured rather than inferred frontier point.

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